

PERSONALITY & EMOTION DETECTION

FROM AUDIO SIGNALS USING DEEP LEARNING

Project Report — PFAI Course

GROUP MEMBERS

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SECTION 1: INTRODUCTION

1.1 Motivation

Human emotion and personality detection from audio signals is one of the most compelling and complex problems in artificial intelligence today. Several key challenges drive the need for robust research in this area.

Challenges

■ Challenge 1: Subjectivity of Emotions

Human emotions are deeply subjective. Two people may express the same emotion in completely different vocal patterns depending on cultural background, age, gender, and individual personality traits, making it extremely difficult to create a universal detection model.

■ Challenge 2: Variability in Audio Quality

Real-world audio recordings suffer from background noise, microphone quality differences, room acoustics, and varying recording conditions. A person recorded in a quiet studio versus a crowded room will produce vastly different audio signals even when expressing identical emotions.

■ Challenge 3: Overlapping Emotional States

Emotions rarely exist in pure, isolated forms. A person can be simultaneously nervous and excited, or sad yet trying to appear happy. These overlapping states create ambiguous audio signals that are extremely hard to classify accurately even for human listeners.

■ Challenge 4: Limited Labeled Audio Data

Unlike image datasets which are abundantly available, labeled emotional audio datasets are scarce. Recording and labeling audio requires significant human effort, domain expertise from psychologists, and careful ethical considerations.

■ Challenge 5: Temporal Nature of Speech

Unlike images which are static, audio is temporal — the emotional content unfolds over time. A sentence spoken quickly in anger differs fundamentally from the same sentence spoken slowly in sadness. Capturing these temporal dependencies requires sophisticated sequential modeling.

■ Challenge 6: Cross-Personality Interference

Personality traits like shyness, introversion, or aggressiveness significantly alter how a person expresses emotions. A shy person expressing anger may sound like a normal person expressing mild frustration, causing misclassification.

Applications

- ✓ **Mental Health Monitoring:** Automated emotion detection can assist psychiatrists in monitoring patients with depression, anxiety disorders, bipolar disorder, and PTSD through voice assistants or mobile apps.
- ✓ **Human-Computer Interaction (HCI):** Smart assistants like Alexa, Siri, and Google Assistant can become significantly more empathetic by understanding the user's emotional state and adjusting responses accordingly.
- ✓ **Customer Service & Call Centers:** Organizations can automatically monitor calls to detect frustrated or distressed customers in real-time, enabling supervisors to intervene immediately.
- ✓ **Education Technology:** E-learning platforms can detect when a student sounds confused, bored, or frustrated and automatically adjust teaching pace, difficulty, or content delivery method.

- ✓ **Automotive Safety:** In-car emotion detection systems can identify when a driver sounds extremely stressed or fatigued and proactively suggest breaks or trigger safety warnings.
- ✓ **Security & Forensics:** Law enforcement agencies can use emotion detection during interrogations to identify stress patterns associated with deception as a supplementary tool.
- ✓ **Social Robotics:** Companion robots designed for elderly care or autism therapy require the ability to understand human emotional states to provide appropriate and safe social interaction.

1.2 Problem Statement

Problem Statement	To develop a deep learning-based system that automatically detects and classifies human personality states (shy, happy, sad, angry, neutral, fearful, disgusted, surprised) from raw audio recordings using acoustic feature extraction and sequential modeling, achieving high classification accuracy on a self-collected and preprocessed audio dataset of minimum 100 samples.
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SECTION 2: BACKGROUND / LITERATURE REVIEW

2.1 Overview

Significant research has been conducted in the domain of Speech Emotion Recognition (SER) and personality detection from audio. Below are four recent and highly relevant works that form the theoretical foundation of this project.

■ Paper 1: "Speech Emotion Recognition Using Deep Learning Techniques" (2021)

Authors: Zhao, J., Mao, X., & Chen, L. | **Published In:** IEEE Transactions on Affective Computing, 2021

Key Contribution: Proposed a dual-branch deep learning architecture that simultaneously processes spectral features (MFCCs) and temporal sequential patterns using both CNNs and LSTM networks, evaluated on RAVDESS and IEMOCAP datasets.

Features	Model	Dataset	Results
MFCC, Chroma, Mel Spectrogram	CNN + LSTM hybrid (dual-branch architecture)	RAVDESS (speech + song), IEMOCAP	RAVDESS: 74.3% IEMOCAP: 68.7%

Relevance: Directly inspired the feature extraction pipeline (MFCC + Mel Spectrogram) and use of a hybrid CNN-LSTM architecture. Demonstrated that combining local feature extraction (CNN) with temporal modeling (LSTM) significantly outperforms either approach alone.

Limitation: Model was trained only on acted/scripted emotional speech, which may not generalize well to spontaneous, natural emotional expressions.

■ Paper 2: "Automatic Personality Perception from Audio-Visual Data" (2022)

Authors: Palmero, C., Sharma, V., Escalante, H.J., et al. | **Published In:** Pattern Recognition Letters, Elsevier, 2022

Key Contribution: Approached personality detection (Big Five: Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism) from multimodal data including audio, video, and text transcriptions.

Features	Model	Dataset	Results
Acoustic Low-Level Descriptors (LLDs), VGGish audio embeddings	Multi-modal fusion network (audio + video + text)	ChaLearn First Impressions Dataset (10,000 video clips)	Audio-only: 61.2% Multi-modal: 73.8%

Relevance: Provided crucial insights into which acoustic features best capture personality-related vocal characteristics. Prosodic features (pitch, speaking rate, voice quality) are strongly correlated with personality traits like extraversion.

Limitation: Reliance on video and text modalities for high accuracy suggests audio-only personality detection remains a challenging open problem.

■ Paper 3: "Attention-Based Bidirectional LSTM for Speech Emotion Recognition" (2022)

Authors: Li, Y., Zhao, T., & Kawahara, T. | **Published In:** Proceedings of Interspeech 2022

Key Contribution: Introduced an attention mechanism on top of Bidirectional LSTM (BiLSTM) networks, allowing the model to focus on the most emotionally salient time frames rather than treating all frames equally.

Features	Model	Dataset	Results
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Log Mel Spectrogram (80 filter banks)	BiLSTM + Self-Attention Layer	EmoDB (German Emotional Speech), CASIA (Chinese)	EmoDB: 85.2% CASIA: 79.6%
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Relevance: The attention mechanism is incorporated into the proposed architecture, enabling the model to identify which parts of the speech signal (e.g., a stressed syllable, a voice break) are most indicative of emotional state.

Limitation: Tested only on single-language datasets; emotional expression varies across cultures and the model may not generalize to other languages.

■ **Paper 4: "Wav2Vec 2.0 for Emotion Recognition" (2023)**

Authors: Pepino, L., Riera, P., & Ferrer, L. | **Published In:** IEEE ICASSP 2023

Key Contribution: Explored self-supervised pre-trained speech representations (Wav2Vec 2.0) for emotion recognition. Instead of manually engineering features, the model learns rich audio representations from raw waveforms through pre-training on massive unlabeled corpora.

Features	Model	Dataset	Results
Raw waveform → Wav2Vec 2.0 embeddings (learned representations)	Transformer-based (Wav2Vec 2.0 + classification head)	IEMOCAP, MSP-Podcast	IEMOCAP: 79.8% (with only 10% labeled data)

Relevance: Motivates the use of pre-trained audio features as a secondary comparison approach. Even with a small dataset (100+ samples), fine-tuning pre-trained representations can yield surprisingly good results.

Limitation: Wav2Vec 2.0 is computationally very expensive; for resource-constrained environments, simpler MFCC + LSTM approaches may be more practical.

Summary Table of Literature Review

Paper	Year	Features Used	Model	Best Acc.	Key Contribution
Zhao et al.	2021	MFCC + Mel Spectrogram	CNN + LSTM	74.3%	Dual-branch hybrid architecture
Palmero et al.	2022	Acoustic LLDs + VGGish	Multimodal Fusion	73.8%	Personality (OCEAN) detection
Li et al.	2022	Log Mel Spectrogram	BiLSTM + Attention	85.2%	Attention mechanism for SER
Pepino et al.	2023	Raw Waveform (Wav2Vec)	Transformer	79.8%	Self-supervised pre-training

SECTION 3: DATASET DETAILS

3.1 Dataset Overview

As per project requirements, preprocessed datasets available online are NOT used. Instead, a raw audio dataset is collected through a structured self-collection methodology.

Attribute	Value
Dataset Name	PEAD — Personality & Emotion Audio Dataset
Total Samples	Minimum 150 audio recordings
Format	WAV (uncompressed, lossless audio)
Sample Rate	44,100 Hz (CD Quality)
Bit Depth	16-bit
Language	English (with some Urdu/bilingual samples)
Collection Period	3–4 weeks
Unique Speakers	30 (18–45 years, 50% male / 50% female)

3.2 Emotion / Personality Classes

Class Label	Description	Target Samples
Happy	Joyful, excited, enthusiastic speech	20
Sad	Sorrowful, melancholic, low-energy speech	20
Angry	Aggressive, loud, fast-paced speech	20
Shy / Nervous	Hesitant, quiet, stuttering, nervous speech	20
Fearful	Trembling voice, rapid speech, anxious tone	15
Disgusted	Contemptuous, repulsed vocal expressions	15
Surprised	Sudden exclamation, pitch spikes	15
Neutral	Flat, emotionless, baseline speech	25
TOTAL		150

3.3 Data Collection Methodology

Step 1: Recruitment of Speakers

30 unique volunteers aged 18–45 years (50% male, 50% female) recruited from university students, friends, family, and social media. Written informed consent obtained; all recordings anonymized by Speaker_ID.

Step 2: Script Design

Participants were given a standardized script with emotionally loaded sentences that naturally elicit the target emotion. Each participant records 5 sentences per assigned emotion class.

Step 3: Recording Setup

Professional (lab-style) recordings (50%): USB Condenser Microphone, quiet room, Audacity software, microphone 15–20 cm from speaker. Naturalistic recordings (50%): smartphone built-in microphone in bedroom, office, and low-wind outdoor environments to introduce realistic variability.

Step 4: Recording Protocol

5-second silence before speaking → state the emotion → speak the script with genuine emotional effort → 5-second silence. Each session lasts 30–45 minutes.

3.4 Raw Dataset Specifications

Attribute	Value
Total Audio Files	150 WAV files
Average Duration / Sample	4–8 seconds
Total Dataset Duration	~15–20 minutes of audio
Total File Size	~500 MB – 1.2 GB (uncompressed)
Unique Speakers	30
Recording Environments	Quiet studio, bedroom, office, outdoor

3.5 Data Quality Control

■ Acceptance Criteria:

- Audio is clearly audible
- Target emotion is genuinely conveyed
- No clipping/distortion
- Background noise is not dominant (SNR > 15 dB)

■ Rejection Criteria:

- Laughing out of character mid-recording
- Heavy background noise (traffic, TV, etc.)
- Recording cuts off prematurely
- Speaker's voice breaks due to laughter (not emotion)

Quality Control Result: Approximately 10–15% of raw recordings are rejected and re-recorded, ensuring final 150 samples meet quality standards.

SECTION 4: PROPOSED METHODOLOGY

4.1 Complete System Pipeline

Stage	Component	Details
1	Dataset Collection	150 WAV files, 8 classes, 30 speakers, professional & naturalistic environments
2	Data Preprocessing	Noise reduction, silence trimming, normalization, fixed-length segmentation
3	Data Augmentation	5× augmentation: noise, time stretch, pitch shift → 150 → 900 samples
4	Feature Extraction	MFCC (+ Δ + $\Delta\Delta$), Mel Spectrogram, Chroma, ZCR, RMS → (134 × T) matrix
5	Model Training	CNN + BiLSTM + Attention, Adam optimizer, 80/10/10 speaker-independent split
6	Evaluation	Accuracy, Precision, Recall, F1-Score, Confusion Matrix, ROC-AUC
7	Post-Processing	Confidence thresholding (60%), sliding-window majority voting, temporal smoothing

4.2 Stage 2: Data Preprocessing

Audio Loading & Resampling: All files resampled to a uniform 22,050 Hz using librosa.load() to ensure consistency across recording devices.

Silence Trimming: Leading/trailing silences (from the 5-second buffer) automatically removed using librosa.effects.trim() with a 20 dB threshold.

Noise Reduction: Spectral subtraction noise reduction applied via the noisereduce library using the first 0.5 seconds as a noise estimate.

Audio Normalization: Peak normalization scales all audio to a maximum amplitude of 1.0, ensuring consistent amplitude across speakers and environments.

Fixed-Length Segmentation: All samples fixed to exactly 5 seconds (110,250 samples at 22,050 Hz) — truncated if longer, zero-padded if shorter.

Data Augmentation (5× factor): Gaussian noise, time stretch ×0.9, time stretch ×1.1, pitch shift +2 semitones, pitch shift −2 semitones → 150 → 900 samples.

4.3 Stage 3: Feature Extraction

Feature extraction converts raw audio waveforms into meaningful numerical representations. Five complementary feature sets are extracted and concatenated:

Feature	Shape	Emotional Relevance
MFCC (40 coefficients + Δ + $\Delta\Delta$)	(120 × T)	Captures vocal tract shape characteristics. Different emotions produce distinctly different vocal tract configurations — angry speech has higher energy across frequencies; shy speech has concentrated low-frequency energy.
Mel Spectrogram (128 filter banks)	(128 × T)	Provides a rich visual-like representation of audio over time. Especially useful for CNN layers that treat it as a 2D image. Captures both frequency content and temporal evolution of emotions.

Chroma Features (12 pitch classes)	(12 × T)	Captures pitch class information. Happy speech tends toward varied, higher-pitched harmonic content; sad speech tends toward monotonic, lower harmonic patterns.
Zero Crossing Rate (ZCR)	(1 × T)	High ZCR indicates noisy, fricative-heavy speech (anger, fear). Low ZCR indicates smooth, voiced speech (sadness, neutrality).
Root Mean Square Energy (RMS)	(1 × T)	Angry speech: high, consistent RMS. Shy/nervous: low and variable. Neutral: steady, medium RMS.

All features are concatenated along the feature dimension to produce a final combined feature matrix of shape **(134 × T)**, where T = number of time frames.

4.4 Stage 4: Model Architecture

A hybrid CNN + Bidirectional LSTM + Attention architecture is proposed:

Layer / Block	Configuration	Purpose
Input Layer	(134 features × T time frames)	Receives feature matrix
CNN Block	Conv1D(64, k=3) → BN → ReLU → Conv1D(128, k=3) → BN → ReLU → MaxPool(2) → Dropout(0.3)	Extract local temporal patterns
BiLSTM Block	BiLSTM(256, return_seq=True) → BiLSTM(128, return_seq=True) → Dropout(0.4)	Capture long-range temporal dependencies (forward & backward)
Attention Layer	Dense(1, tanh) → Softmax → Multiply → Sum	Focus on most emotionally salient time frames
FC Layers	Dense(256) → BN → ReLU → Dropout(0.5) → Dense(128) → Dropout(0.3)	High-level feature transformation
Output Layer	Dense(8) → Softmax	Probability distribution over 8 emotion classes

4.5 Stage 5: Model Training & Evaluation

Split	Samples	Percentage	Note
Training Set	720	80%	Speaker-independent split
Validation Set	90	10%	Used for early stopping
Test Set	90	10%	Final evaluation only
Total	900	100%	After 5× augmentation

Training configuration: Adam optimizer (lr=0.001), categorical cross-entropy loss, 100 epochs, batch size 32, early stopping (patience=15), ReduceLROnPlateau (patience=7), ModelCheckpoint (save best weights).

Important: The split is speaker-independent — recordings from the same speaker never appear in both training and test sets, preventing the model from recognizing a speaker's voice instead of learning emotional patterns.

Evaluation Metrics

- Accuracy: Overall percentage of correct predictions
- Precision, Recall, F1-Score: Per-class and weighted average metrics

- Confusion Matrix: Visual representation of classification results across all 8 classes
- ROC-AUC Curve: Per-class discrimination ability

4.6 Stage 7: Post-Processing

Confidence Thresholding (60%): If the model's maximum class probability is below 60%, the output is flagged as "UNCERTAIN" rather than making a potentially incorrect high-stakes prediction.

Sliding Window Majority Voting: For audio clips longer than 5 seconds, the audio is divided into overlapping 5-second windows (2-second hop) and the final prediction is determined by majority voting across all windows.

Temporal Smoothing: For real-time applications, predictions from consecutive windows are smoothed using a moving average (window=5) to prevent rapid, jittery label switching.

Summary Table — Pipeline Stages

Pipeline Stage	Tools / Libraries	Input	Output
Data Collection	Audacity, Smartphone	Human speech	150 WAV files
Preprocessing	Librosa, Noisereduce	Raw WAV	Clean, normalized WAV
Augmentation	Librosa	150 samples	900 samples
Feature Extraction	Librosa, NumPy	Audio waveform	Feature matrix (134×T)
Model Training	TensorFlow 2.x / Keras	Feature matrices	Trained model (.h5)
Evaluation	Scikit-learn	Test features	Accuracy, F1, Confusion matrix
Post-Processing	NumPy, SciPy	Raw probabilities	Smoothed, thresholded prediction
Output	Custom visualization	Probabilities	Emotion label + confidence score

Tools & Technologies Summary

Category	Tool	Purpose
Recording	Audacity, Smartphone	Audio data collection
Audio Processing	Librosa	Feature extraction, augmentation
Noise Reduction	Noisereduce	Preprocessing
ML Framework	TensorFlow 2.x / Keras	Model building & training
Data Handling	NumPy, Pandas	Data manipulation
Visualization	Matplotlib, Seaborn	Plotting features, confusion matrix
Evaluation	Scikit-learn	Metrics computation
Environment	Jupyter Notebook / Google Colab	Development environment